Close to the cap: who plays the whole match across nine sports

Abstract

Sports governing bodies keep changing the number of substitutions a team may make during a match. We call this number the cap. We asked how close teams come to their cap. We read the rulebooks and match records of nine sports side by side. The records hold over a million appearances. From the rulebook alone, a governing body can work out the largest share of starters a new cap lets a team take off. Where a replaced player cannot return, teams used most of their cap and came close to that share under the caps they had. Where players may return, teams with a cap also used most of it. Yet under caps of ten or fewer, most of their starters still played the whole match. Simple counting also gives data providers a free check on their match records. For match simulators, we built a model of when teams make their substitutions. On unseen matches, all six predictions we wrote down beforehand came true.

Keywords: substitution rules; sports rules; match records; order statistics; underdispersion; data quality.

1 Introduction

Sports governing bodies keep changing the number of substitutions a team may make during a match. We call this number the *cap*. Rugby union calls its substitutes replacements. Rugby league and Australian rules call each substitution an interchange. Football raised its cap from three substitutions to five (The International Football Association Board, 2020, 2022). Rugby union raised its cap from seven replacements to eight. Rugby league cut its cap from ten interchanges to eight. Australian rules cut its cap twice, to 75 interchanges. In football, rugby union, Gaelic football, hurling and baseball, a replaced player cannot return. We call these *one-way sports*. Table 1 lists their few exceptions, such as replacements for blood injuries. In rugby league, Australian rules, ice hockey and basketball, a player may return. We call these four *re-entry sports*. There, one player may come off and go back on several times. So in rugby league and Australian rules, the cap limits substitutions rather than players. Ice hockey and basketball let teams make substitutions freely.

A cap works like a speed limit. Such a limit caps how far a car can go in an hour. The driver chooses the speed below it. In the same way, a cap limits how many starters a team can take off. The coach chooses who comes off. So from the rulebook alone, a governing body can work out the largest share of starters a new cap lets a team take off. Where a replaced player cannot return, teams used most of their cap and came close to that share. Under three substitutions, football's teams took off 26 per cent of their starters, and the cap allows 27.3 per cent. Under five, they took off 39 per cent, and the cap allows 45.5 per cent. Where players may return, teams with a cap also used most of it. Yet under caps of ten or fewer, they took off far fewer starters than the cap allows. Under eight interchanges, 63 per cent of rugby league's starters played the whole match. That is about as many as

football's 60 per cent under five substitutions.

Where a replaced player cannot return, simple counting also ties the starters who leave to the substitutions a team makes. Each substitution takes off either a starter or a player who came on earlier. So the starters who leave equal the team's substitutions, less the substitutions that take off an earlier substitute, plus the starters sent off. An equation that holds by counting alone, like this one, is called an *identity*. It gives governing bodies a formula. It also gives data providers a free check, because a record whose substitutions and minutes disagree is inconsistent. This matters, because rule changes for player safety have been judged on public media records (Beaudouin et al., 2019).

What this means in practice. The results serve four groups: governing bodies, clubs, data providers and builders of match simulators.

- *For governing bodies.* Where a replaced player cannot return, teams use most of their cap, and coaches rarely replace an earlier substitute. Where players may return, teams with a cap also use most of it. Yet under caps of ten or fewer, they take off far fewer starters than the cap allows. In rugby league most starters still play the whole match under its eight interchanges.

- *For clubs.* In football and rugby league, who plays the whole match depends more on position than on the cap. As expected, football's goalkeepers and rugby league's backs play the whole match far more often than outfield starters and forwards. So clubs manage these players' load between matches.

- *For data providers.* Anyone who learns from match data needs records that say who really played the whole match (Pappalardo et al., 2019; Wang et al., 2024). In nine football leagues of one large public dataset, the free check flags 8 to 27 per cent of the teams'

records. Each flagged record credits a starter taken off at minute 90 or later with the whole match. In football the starters taken off and the substitutions made stay within 3 per cent of the team in each month of the season. The rugby league and Australian rules records pass a second check on average: their minutes sum to the players on the field times the length of the match.

- *For simulators.* A match simulator needs to know how many substitutions teams make and when they make them (Titman et al., 2015; Maia et al., 2026). A curve cut off at the cap describes the number of substitutions a team makes. Its width is about one to two substitutions, and its centre moves with the cap. A team's substitutions come, on average, near where the same number of random moments in one stretch of the match would fall. In football, teams make their substitutions while play is stopped. We call each such pause a stoppage. A model of the stoppages met all six pass marks that we wrote down before we read the matches kept back to test it.

What is new. Earlier work has studied substitutions mostly as a coaching decision. It has modelled the best moments to make substitutions (Hirotsu & Wright, 2002, 2005; Hirotsu et al., 2024), timed them (Del Corral et al., 2008; Myers, 2012; Silva & Swartz, 2016; Hills et al., 2019; Sun et al., 2024) and related them to the score (Gómez et al., 2016; Wittkugel et al., 2022). It has measured how their use grew with each change of rule (Nevill et al., 2026; Dilger & Vischer, 2024), how often teams use their whole cap (Ribeiro et al., 2020; Meyer & Klatt, 2021; Roberts et al., 2024; World Rugby, 2020) and how often they make two substitutions at once (Xiao & Zhang, 2024). It has also compared rugby league's whole-match and rotated players (Black & Gabbett, 2014; LeagueEyeTest, 2021), and the caps of different sports

(Mota et al., 2020). This paper asks how close teams come to the cap itself. It studies caps already in force (Wright, 2014). Ricou (2026) first reported, sport by sport, how many starters play the whole match and how closely the starters taken off track the substitutions made. This paper adds how far each league sits above the least share of starters its cap leaves on the field at the final whistle. It also reads who plays the whole match position by position, and it checks match records with the same counting. And it tests a model of when teams make their substitutions on matches kept apart for the test.

2 Rules and records

Table 1 gives each sport's cap and the records we read. The Electronic Companion, the paper's online appendix, lists the key terms and the panel behind each result. We write n_f for the players a team starts with, and $n_{\max}$ for the players a team may bring on. In a re-entry sport the rule limits interchange events rather than players, and we write that event limit n_e. Some matches allow extra replacements, so we also use the *effective limit* n_{eff}, the cap that applies in that match.

Every record comes from public match data. Football's events and appearances come from a public dataset built from Transfermarkt's match pages (Cariboo, 2026). The other sports' records come from the leagues' own pages and statistics interfaces (MLB Advanced Media, 2026; National Hockey League, 2026; Australian Football League, 2026; AFL Tables, 2026; National Rugby League, 2026; It's Rugby, 2026; Euroleague Basketball, 2026; Gaelic Athletic Association, 2026). We call the records read for one analysis its *panel*. We read each panel one team-match at a time, and a *team-match* is one team's record of one match. The three-substitution panels leave out the matches played under the temporary five-substitution amendment of 8 May to 31 July 2020. The count and timing panels leave out the 2019–20

Table 1: The rules and the records. The cap is $n_{\max}$ in the one-way sports and the event limit n_e in rugby league and Australian rules. 'Not counted' lists the replacements the rule leaves outside the cap. The Electronic Companion gives each rule with its reference and explains season names.

Sport	n_f	Cap	Not counted	Seasons	Panel and source	Team-matches
Association football	11	3; 5 from 2020 by the competition's choice, in at most three stoppages in play; +1 in extra time	concussion from 2020	2012–25	29 men's leagues, Transfermarkt; event records name the player replaced in nine	59,982 (five); 58,328 (three); 42,303 (identity)
women				2024	seven leagues, Soccerdonna and FBref; the month-by-month identity only	1,996
Rugby union	15	7, then 8 by league from 2008; 8 in the Laws from 2014; up to four returns in the French leagues, 2018–25	blood, head injury	2004–26	Five leagues, itsrugby.fr; the identity also on the 2025–26 Top 14 sheets	25,028 (identity); 25,852 (count)
Gaelic football	15	5 to 2013, 6 in 2014–19, 5 from 2020; +3 in extra time	blood, head injury	2002–24	All-Ireland senior finals, match reports	10
Hurling	15	5; +3 in extra time	blood, head injury	2003–25	All-Ireland senior finals, match reports	18
Baseball	10	none; at most 13 pitchers from 2022; a pitcher faces three batters from 2020	–	2008–25	MLB Stats API game logs; the identity on 2024	77,096 (count); 4,858 (2024)
Rugby league	13	10 to 2015, 8 from 2016; bench 4	head injury, 18th player	2013–25	NRL match-centre feed	5,061
Australian rules	18	none to 2013, 120 in 2014–15, 90 in 2016–20, 75 from 2021; bench 4, 5	blood, head injury, changes at the breaks, the substitute	2013–25	AFL Tables and the AFL's statistics API, men and women	2,916 (men); 198 (women)
Ice hockey	6	none; 20 dressed	–	2008–24	NHL game sheets and shift charts, public statistics API	2,080 (2023); 730,501 skater-games
Basketball	5	none; 12 dressed	–	2007–24	EuroLeague and EuroCup box scores, 29 competition-seasons	10,830

season altogether. Its matches under the amendment are kept apart to test the model of Section 5, and we call them the *held-out matches*.

In the re-entry sports, the minutes t_i of a team's players should add up to the players on the field times the length of the match t_M, so that $\sum_i t_i = n_f\, t_M$. Injury researchers call this total *exposure* (Fuller et al., 2006). Each of 10 panels in two sports passes this check on average. In the Australian Football League (AFL), the top competition of Australian rules, the sum is within 0.03 per cent of that total.

3 Who plays the whole match

Where a replaced player cannot return, teams used most of their cap and came close to the largest share of starters it lets them take off. Where players may return, teams with a cap also used most of it. Yet under caps of ten or fewer, they took off far fewer starters than the cap allows. We say that a starter plays the whole match when the starter is on the field from the start to the final whistle.

3.1 The most a cap lets a team take off

Where a replaced player cannot return, each substitution takes off either a starter or a substitute who came on earlier. We call the second kind of substitution a *chain*. So each of the other substitutions takes off one starter. A starter recorded as leaving before the end of the match is *withdrawn*, and a team's *withdrawal rate* s_w is the share of its starters withdrawn.

Proposition 1. *In a one-way sport, let a team-match have S substitutions, of which C are chains, L take off a starter whom the record credits with the full match, and X take off a player on neither the starters' nor the substitutes' list. Let D be the starters sent off and not*

replaced, and U the other starters recorded short with no substitution or card. Then

$$n_f \left(\frac{S}{n_f} - s_w \right) \;=\; C + L + X - D - U, \tag{1}$$

and, because $S \leq n_{\text{eff}}$,

$$s_w \;\leq\; \frac{n_{\text{eff}} + D + U}{n_f}. \tag{2}$$

The proof is only counting, and the Electronic Companion writes it out. In words, a team's substitution rate S/n_f, its substitutions per player on the field, equals its withdrawal rate apart from five corrections. Of the five, only chains are a coach's choice. A *late entry* is a substitution made at minute 90 or later, in the added time at the end of the match. The record lists it at minute 90, so its starter seems to finish. Late entries and players on neither list come from how the record is kept. Starters sent off and other exits without a replacement are *departures without replacement*. They are events of the match, and the record gets them right. Take a football team that makes five substitutions. One of them takes off a player who had come on earlier, so it is a chain. Another comes in added time, and the record lists its starter as finishing the match. The record then shows three of its 11 starters withdrawn, two fewer than its substitutions. Each of the two missing starters is one of the five corrections.

The second equation is the bound the cap sets. Its substitution part, n_{eff}/n_f, needs only the rulebook. So at most n_{eff}/n_f of the starters can be taken off by substitution, and at least $1 - n_{\text{eff}}/n_f$ play the whole match, less departures without replacement. We call this least share the *floor*. A football team allowed five substitutions can take off at most five of its eleven starters, 45.5 per cent, so at least 54.5 per cent play the whole match.

3.2 Where a replaced player cannot return

Where a replaced player cannot return, the share of starters who played the whole match sat just above the floor (Figure 1a). In football it sat 0.4 percentage points above it under three substitutions and 5.3 under five. In rugby union it sat 3.3 to 5.6 points above it. There, with a bench of eight, about half the starters played the whole match. In the All-Ireland finals of Gaelic football and hurling, 75 and 71 per cent did. So, as the floor guarantees, about half or more of the starters played the whole match under each of these caps.

Teams came this close because they used their caps. In football most teams used the whole cap: 85 per cent did under three substitutions and 56 per cent under five. And coaches rarely took off a substitute. Football's event records name, for every substitution, the player who left and the player who came on. So in nine leagues every part of equation (1) can be counted, on 42,303 team-matches. We call the share of substitutions that are chains the *chain share* $s_c = C/S$. In these leagues it is only 0.2 to 0.9 per cent. So the starters taken off follow the substitutions almost one for one. We call one league under one substitution rule a *league-regime*. Figure 2 places every league-regime with the substitution rate across and the withdrawal rate up. We call the difference between the two rates the *gap*, and give it in per cent of the team. Every one of football's 18 league-regimes has a gap below 3 per cent. That is 0.33 of a starter per team-match. We call this level the *tolerance*. Rugby union's lineups, which give both counts from one list, show a gap of 0.2 per cent. The gap also stays small in each month of the season. Its largest monthly value in football's European leagues is 2.3 per cent, inside the tolerance.

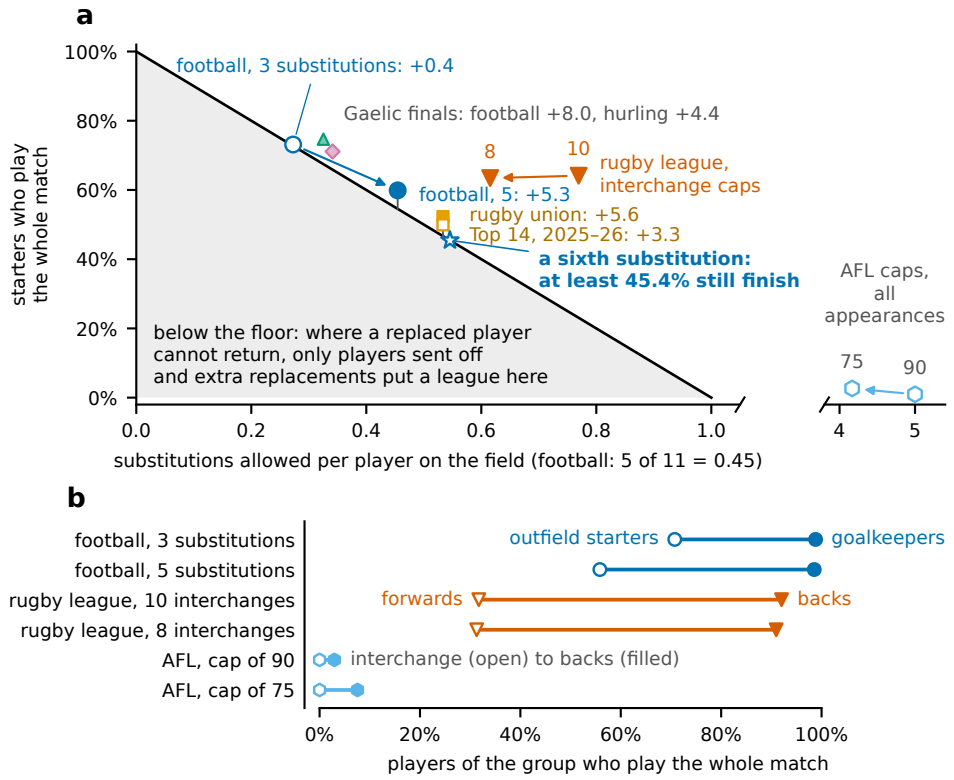


Figure 1: A cap sets a floor under the starters who finish, and the leagues where a replaced player cannot return sit close above it. (a) Where a replaced player cannot return, at least $1 - n_{\text{eff}}/n_f$ of the starters finish, less those sent off: the line is this floor, and the region below it is shaded. Labels and thin grey lines give each league's distance above it in percentage points; the Gaelic finals are plotted at five substitutes, though the 2015 football final allowed six. Football is read at the final whistle, and rugby union's open square is the Top 14 on its newest lineups. Arrows join each change of cap. Rugby league and the AFL let players return, so their caps count interchanges; the AFL sits in a compressed segment. (b) Under each cap, the group of positions that plays the whole match most often (filled) and least often (open). In football and rugby league the groups differ more than a change of cap moved them, and in the AFL almost every player comes off.

Alt text: Panel a: a falling floor line over a shaded region; football's and rugby union's points sit just above it, the lighter Gaelic points close above it too, a star on the floor, rugby league's points higher to the right and the AFL's near zero. Panel b: six bars between two markers, long for football and rugby league, short near zero for the AFL.

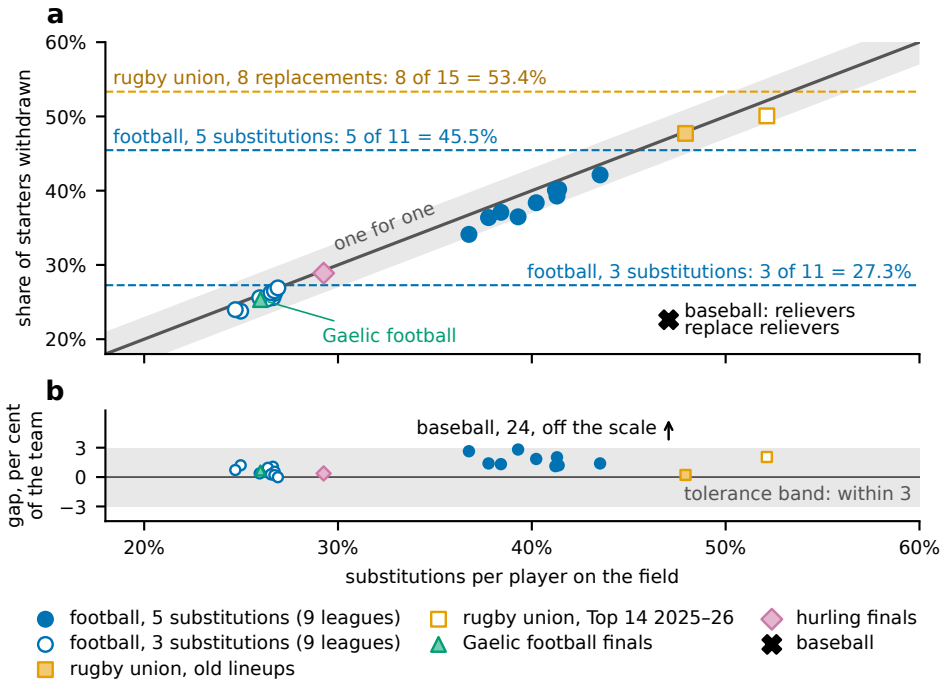


Figure 2: Substitutions take off starters almost one for one. (a) Each point is one league under one substitution rule, a league-regime, with its substitutions per player on the field across and the share of its starters withdrawn up. The diagonal is one starter withdrawn per substitution, and the grey tolerance band spans 3 per cent of the team either side of it. Football's filled points are under five substitutions and its open points under three, and each dashed line is the ceiling a cap sets, its share rounded up. Rugby union's old lineups and the Gaelic finals give both counts from one list, so their agreement is close to built in (lighter markers). The open square is France's Top 14 in 2025–26, on the newest lineups, which list every player's minutes. Baseball lies far below the band, because at least 45 per cent of its substitutions take off a player who had come on earlier, a chain. (b) Each gap, the substitution rate less the withdrawal rate, in per cent of the team, with the same band. Figure EC1 draws each gap with its 95 per cent interval, which shows how precise it is.

Alt text: Panel a: a scatter with a diagonal line inside a grey band. The eighteen football points lie in the band below their dashed cap lines, with the rugby union, Gaelic football and hurling points, and one baseball point sits far below it. Panel b: the same points as gaps near zero inside a narrow band, with an arrow for baseball off the top.

3.3 Where players may return

Where players may return, most starters still played the whole match under caps of ten or fewer. In the National Rugby League (NRL), the top rugby league competition of Australia and New Zealand, a player may return. Under eight interchanges, 63 per cent of its starters played the whole match. That is about as many as football's 60 per cent under five substitutions. The NRL's share sits 25 points above its floor. So its teams took off far fewer starters than the cap allows. Yet teams used 89 to 96 per cent of their cap under each of five caps where players may return. As expected, teams spent the interchanges on the forwards. The backs lasted the whole match in 91 per cent of their appearances under eight interchanges, and the forwards in 31 per cent (Figure 1b). The cut from ten interchanges to eight moved each group by 1.1 percentage points or less. Under a cap several times the size of the team, as in Australian rules, almost everyone comes off. Only 1 to 3 per cent of its appearances last the whole match. Where the rules set no cap, as in ice hockey and basketball, almost every player comes off at some point.

In football too, positions differ more than a change of cap moved them. Its goalkeepers finished 99 per cent of matches under both caps, and its outfield starters went from 71 to 56 per cent. So in football and rugby league, position matters more than the cap.

3.4 Checking the records

Where a replaced player cannot return, the same counting gives a free check on match records. On records that name the player replaced, every part of the identity can be counted. We flag a team's record when it has at least one late entry or one player on neither list. A flag needs only the record's list of substitutions and its players' minutes. The check flags 8 to 27 per cent of the teams' records here. Every flag

is a late entry, because the source lists added time at minute 90. Per team-match the flags rose in every league when teams could make five substitutions (Figure 3a). Most of the gap is late entries. Chains add very little. Departures without replacement pull the gap the other way (Figure 3b).

4 How many substitutions a team makes

A narrow count. The number of substitutions a team makes varies little. Its standard deviation, the typical distance from the average, is only 0.44 to 1.25 substitutions. In football most teams use the whole cap (Section 3). The standard model of random match events is a Poisson count (Maher, 1982; Dixon & Coles, 1997). It spreads teams far too widely, and in football it puts 11.1 per cent of teams in the wrong count (Figure 4).

A narrow curve, cut off at the cap. We summarize the count with a normal distribution, the bell-shaped curve, with centre μ and standard deviation σ. We round it to whole numbers and cut it off at the cap, and we call this curve the *censored normal*. For each sport we fit it, choosing the centre and width that match the data best. Its centre and width describe the curve as a whole. It puts only 0.6 per cent of football's teams in the wrong count, and at most 4.2 per cent in the other count panels. Its width, its standard deviation before the cap cuts it, is 0.95 to 1.76 substitutions in every count panel. Its centre moves with the cap. In rugby union the centre is 2.20 replacements higher in the seasons with a bench of eight than in the earlier seasons with seven. It can move by more than one replacement, because it is the centre of the curve before the cut.

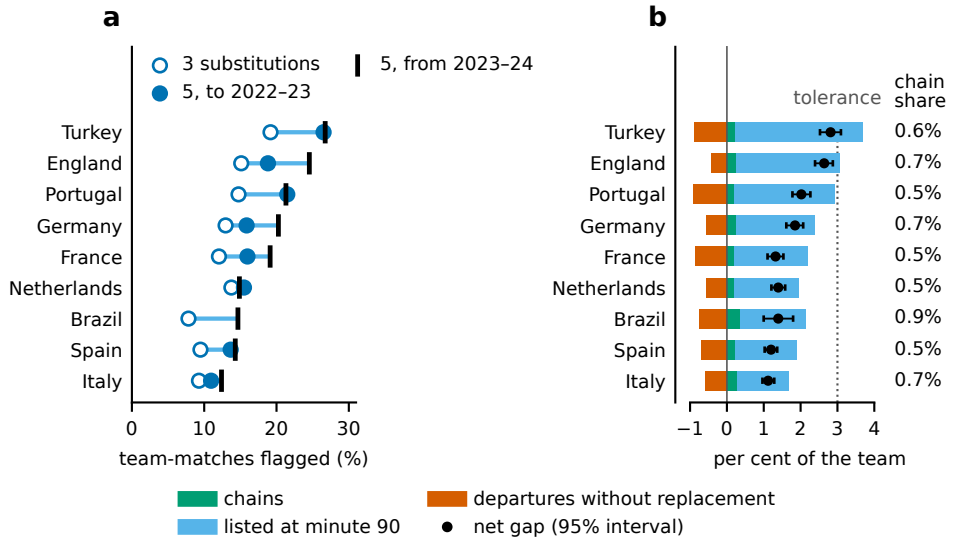


Figure 3: Counting flags 8 to 27 per cent of football's team-matches, every flag a substitution listed at minute 90. Per team-match the flags rose in every league at the move to five substitutions, and in England and Germany most of the rise came from 2023–24. In England that was the season the Premier League began to add the exact time lost. (a) The share of team-matches flagged in each league, pooled over its seasons under three substitutions (open), under five before 2023–24 (filled) and under five from 2023–24 (black tick), joined in that order; a team-match is flagged when its record credits a starter taken off by a substitution with the whole match. Brazil's only season under five substitutions is 2024, so it has a tick and no filled circle. The two rules' seasons are different years. (b) Football's gap under five substitutions, part by part, in per cent of the team and in the rows of (a): chains, the substitutions that take off a player who had come on earlier; substitutions listed at minute 90; and, to the left, departures without replacement, starters sent off or leaving unreplaced. By Proposition 1 the parts sum to the gap exactly. The black mark is the net gap with its 95 per cent interval, the dotted line its tolerance, and the right column the chain share, the chains' share of the substitutions.

Alt text: Panel a: for nine football leagues, a line runs from an open circle through a filled circle to a black tick; every filled circle lies right of its open circle, and the tick lies furthest right of the filled circle in England and Germany and close to it in the Netherlands, Portugal and Turkey. Panel b: for the same leagues, horizontal bars made mostly of substitutions listed at minute 90, with a thin green sliver of chains, a shorter bar to the left for departures without replacement, a black dot for each net gap and a dotted line for the tolerance to the right of every dot.

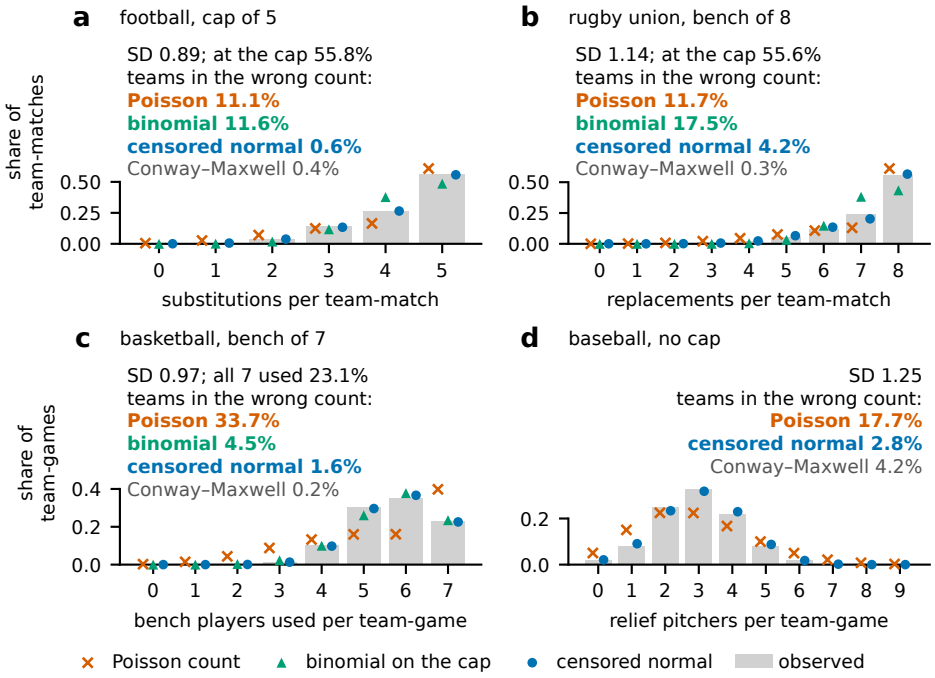


Figure 4: The number of substitutions varies little: the censored normal comes close to its shares, and a Poisson count does not. Grey bars are the observed shares, crosses the best-fitting Poisson count, triangles the binomial count on the cap with the observed mean (none in d), and dots the censored normal, each cut off at the cap. Each panel prints the observed standard deviation (SD), the share of teams at the cap (in basketball, the share that use all seven bench players), and the share of teams each model puts in the wrong count, that is, the share it would have to move to match the observed shares; its last line is the best Conway–Maxwell family, a standard model for counts narrower than Poisson, which is not drawn and which Figure EC3 compares. The Poisson count spreads teams too widely, and in football and rugby union the binomial is too narrow, because many teams use the whole cap and some use only a few substitutions. (a) Football under five substitutions. (b) Rugby union's sides with a bench of eight. (c) Basketball's bench players used, of the seven on its bench. (d) Baseball's relief pitchers, where no cap binds, shown to nine.

Alt text: Four bar charts of the number of substitutions per team. Blue dots sit on top of the grey bars. Red crosses miss them, too high at low counts and at the cap. Green triangles pile too much weight one substitution short of the cap in football and rugby union.

5 When the substitutions come

Timing read from the count. On average, a team's substitutions come where the same number of random moments in one stretch of the match would fall. Among the teams that make exactly n substitutions, the k-th substitution has an average minute. We call that average a *cell*. Now picture n moments dropped at random on a stretch of the match and put in order, which statisticians call order statistics. On average the k-th of them falls at the fraction $k/(n+1)$ of the stretch, a classical result of probability (Cunnane, 1978; David & Nagaraja, 2003). A team that makes three substitutions, for example, would put the first a quarter of the way through the stretch. It would put the second halfway and the third three quarters of the way. We call the stretch of the match in which a team's substitutions fall its *window*. If a team's substitutions behave like such random moments on a window, the cells fall on the line

$$\mathrm{E}[t_{k:n}] \;=\; t_M \left(\zeta_{0,n} + \zeta_1 \, \frac{k}{n+1} \right). \tag{3}$$

Here $\zeta_{0,n}$ is where the window opens for a team with n substitutions, and ζ_1 is how wide it is. Both are shares of the match. We call this the order-statistic line.

The line summarizes the averages. Figure 5a shows every cell of football's five-substitution era and of rugby union, and the line passes close to almost every cell. It misses the cells' average minutes by 1.1 minutes in football and 1.3 in rugby union. Putting every substitution at the average minute would miss by 8.7 and 7.8 (Figure 5c). With an opening for each number of substitutions, the line shows that teams that make more substitutions start earlier. Their window opens at 51.3 minutes for three substitutions, 50.8 for four and 48.5 for five.

Stoppages, and the bundling they require. The line gives the average minute of each substitution, but a simulator needs the substitutions of whole matches. Minutes picked at random, one by one, put two consecutive substitutions in one minute 6 per cent of the time. In the records 37 per cent share a minute. The reason is that a team using five substitutes may make them in at most three pauses during play. We call each pause in which a team makes one or more substitutions a *stoppage*.

Proposition 2. *Let a team stop play for substitutions at most w times during play, besides half-time, and let it make n substitutions, h of them at half-time. Then at least* $\max(0,\, h-1) + \max(0,\, n-h-w)$ *of its* $n-1$ *consecutive pairs of substitutions share a stoppage.*

Half-time holds its h substitutions in one stoppage, and the $n-h$ substitutions made during play fall in at most w stoppages. So the fewer stoppages a team may use, the more of its substitutions must share one. With w at three and each team's own h, football's rule forces 35.0 per cent of consecutive substitutions into one stoppage. So the rule forces most of the bunching in the records.

A model of the stoppages. We built a model of the stoppages in three steps from football's five-substitution seasons, which we call the *fitted seasons*. A stoppage here is the substitutions listed in one minute. First, a team's number of substitutions is picked at random from the censored normal of Section 4. Second, the team splits its substitutions between half-time and its stoppages in play, in the proportions that teams with the same number of substitutions used. Third, a team with m stoppages in play picks their minutes at random from the minutes at which teams with m stoppages in play stopped play. It picks again if two minutes are the same, and puts them in order. So its j-th stoppage

in play comes at the j-th of m ordered random minutes

$$\tau_{j:m}, \qquad \tau_1, \ldots, \tau_m \text{ drawn independently from } F_m, \text{ a set with a repeated minute being redrawn.} \tag{4}$$

Here F_m is the distribution of the minutes at which teams with m stoppages in play stopped play. Each substitution then takes its stoppage's minute. The order-statistic line is the case in which every substitution is its own stoppage and F_m is spread evenly over the window. On the fitted seasons the model misses the cells' average minutes by 0.6 minutes, against 1.1 for the line.

We then tested the model on the held-out matches, which this paper had kept apart until then. They are the 1,270 team-matches of the 2019–20 season played under the temporary amendment of May 2020. We wrote down six predictions with their pass marks in a register before reading them, and every one held (Electronic Companion). The model missed the cells' average minutes by 0.9 minutes, against 1.5 for the line. It put 37 per cent of consecutive substitutions in one minute, against 33 per cent in these matches' records (Figure 5b).

6 Discussion

6.1 Managerial implications

For governing bodies. The counting gives governing bodies a simple formula. A cap of n_{eff} limits the share of starters a team can take off by substitution to n_{eff}/n_f, and so it sets the floor of Section 3.1. Figure 1a turns this formula into a design chart. Where a replaced player cannot return, teams came close to the floor under the caps they had (Section 3). Where players may return under caps of ten or fewer, the formula guides a cap only loosely. Rugby league's starters

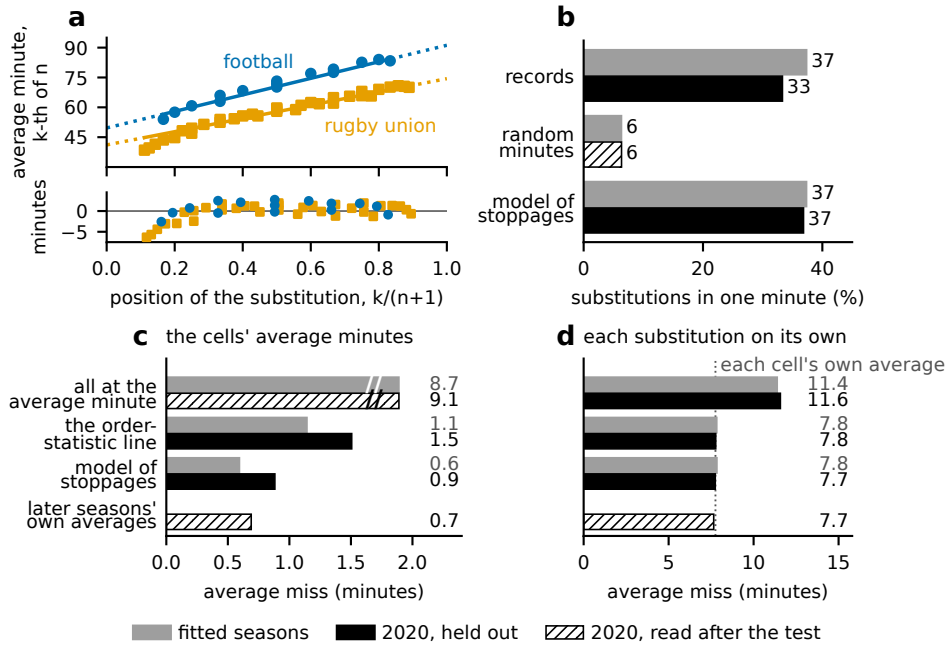


Figure 5: The average minute of each substitution can be read from the number of substitutions, and substitutions come bundled in stoppages. (a) Each point, a cell, is the average minute of the k-th of n substitutions in football's five-substitution era and in rugby union, with the order-statistic line, drawn here with one opening shared by all n, dotted past the cells; the strip gives each cell's distance from it in minutes. First substitutions come early, partly for injuries. Rugby union's cells sit lower because its match lasts 80 minutes. (b) Consecutive substitutions made in one minute: in the records, for minutes picked at random and for the model of stoppages, which is built to match the fitted seasons. (c) How far each placement misses football's cell averages, each cell weighted by its substitutions, on the fitted seasons' 15 cells and the 14 held-out cells with at least 50 substitutions; the average minute's bars are cut. The later seasons' own averages, with no grey bar as the fitted seasons are their source, miss slightly less than the model, a difference whose 95 per cent interval includes zero. (d) The same for each substitution on its own; the dotted line is the miss of each cell's own average on the fitted seasons, 7.8 minutes. In (b) and (c) the black bars answer predictions registered before the held-out matches were read; hatched bars were read after the test.

Alt text: Panel a: points along two rising lines, football above rugby union, most within two minutes of their line. Panels b to d: bars, short in b for random minutes, cut in c for the average minute, near eight minutes in d.

sat 25 points above its floor. How close teams would come to a new cap is a forecast. A sixth substitution, for example, would raise football's ceiling from 45.5 to 54.6 per cent of the team. At least 45.4 per cent of the starters would still finish. The Gaelic Athletic Association's football advisory group has proposed a sixth substitute (Harrington, 2026). It would lower Gaelic football's floor from two thirds of the starters to 60 per cent. Proposition 2 does the same for stoppages. With today's numbers of substitutions, two stoppages in play would force at least 62 per cent of consecutive substitutions into one stoppage. Four would force at least 14 per cent. Every rulebook here that sets a cap now allows replacements for head injuries outside it. So the formula also shows how many more starters each exemption lets a team take off. The formula is one input among several, beside player welfare, tactics, fairness, the flow of the match, spectators, heat and cost (Electronic Companion). It does not rank these goals, but it tells a governing body exactly how far each option moves the share of starters who can be withdrawn.

For statistics units and data providers. In sports where no player may return even for a time, the identity gives a check at two levels. On records that name the player replaced, a data provider can flag each team-match with a late entry or a player on neither list. Where records do not pair players, as in most public appearance tables, the gap checks them across many matches once the starters sent off are added back. A sum near zero then points to substitutions missing from the record, as in a first reading of Brazil's top league (Electronic Companion). Such a source's minutes also stop at minute 90 for every player, so injury research must add the added time back to measure exposure. A provider that keeps added time in its minutes, as Wyscout does (Wyscout, 2026), raises no flag.

For builders of match simulators. A simulator can pick the number of substitutions at random from a censored normal. It can then pick the team's stoppages as equation (4) describes, and give each substitution its stoppage's minute. On the held-out matches, the censored normal gave 4.9 per cent of teams the wrong number of substitutions. A Poisson count gave 13 per cent. The model put 37 per cent of consecutive substitutions in one minute. Minutes picked at random put 6 per cent there, and the held-out records 33 per cent. For each substitution on its own, the model and the line miss alike (Figure 5d), so the model's gain is in how it groups substitutions. The Electronic Companion's tables let a simulator run the model.

What the paper does not do. The paper makes no forecast of how teams would respond to a new cap. It also draws no link to results, because both need study designs this paper does not have.

6.2 Limitations

First, the agreement between the two rates has a boundary. Baseball is strictly one-way, yet its gap is 24 per cent, because each relief pitcher after the first replaces a pitcher who had come on as a substitute. So a small gap needs few chains as well as a ban on re-entry.

Second, the tolerance of 3 per cent was set on football after football's gaps had been read, so the gaps are descriptions, not tests. The predictions for the other sports were entered in the register only after their tests had been written. A registered ordering of who plays the whole match, by the cap per player on the field, failed in rugby league. So the line between caps of ten or fewer and very large caps was drawn after that test.

Third, rugby union's gap includes temporary replacements, the sin-bin and the front-row replacement rules. The Gaelic panels hold only

the finals with complete and consistent lineups. The AFL's own interchange records may make its exposure check pass automatically.

Fourth, of the 39 predictions in the paper's main register, nine failed and three could not be checked (Electronic Companion).

Fifth, the count models are summaries rather than laws. In rugby union and basketball a Conway–Maxwell family, a standard model for counts narrower than Poisson, fits better than the censored normal (Shmueli et al., 2005; Kadane, 2016).

Sixth, the window describes the middle of the sequence, not its ends, and in rugby union even the middle substitutions reject one window. The model of stoppages was tested on one short period under the same rule, and its pass marks were set after the fitted seasons had been read. After the test, the later fitted seasons' own cell averages missed slightly less than the model, and copying whole sequences from real matches did better still. So four predictions are registered for the 2026–27 seasons, the first under the ten-second exit. This study ends before those seasons end, so it does not read them.

Seventh, the comparisons across rules are descriptions, not effects, because football's three- and five-substitution seasons are different years. Comparisons between sports are descriptions too, because the sports differ in more rules than the cap, and their clubs differ in their habits.

Last, the analyses use different samples, and caps are themselves outcomes of rule-making (Australian Football League, 2013).

7 Conclusion

The cap works as a speed limit does. It caps how many starters a team can take off, and the coach decides whom to keep on. Where a replaced player cannot return, teams came close to that cap. Where players may return, most starters still played the whole match under

caps of ten or fewer. As expected, in football and rugby league, who plays the whole match depends more on position than on the cap. Simple counting gives governing bodies a formula and data providers a free check on match records. It does not say how teams will respond to a new cap. The number of substitutions and their timing follow regularities that simulators can use. The model of stoppages met its registered predictions on matches it had not seen.

Data availability

The deposits behind every number in this paper, the software that produces them, and a ledger that maps each printed number to its deposit will be archived with the paper in a public repository. No login, licence or fee was involved, and the deposits hold only aggregates, never a named player. The fits use SciPy (Virtanen et al., 2020) and the figures Matplotlib (Hunter, 2007). The archive identifier is withheld for anonymized review.

References

AFL Tables (2026), ‘AFL tables’, Match statistics site, read 2026.
`https://afltables.com/afl/afl_index.html`

Australian Football League (2013), ‘AFL commission introduces 120 limit on interchanges’, News release, 27 August.
`https://www.afl.com.au/news/452935/afl-commission-introduces-120-limit-on-interchanges`

Australian Football League (2026), ‘AFL match-centre statistics API’, Public statistics interface, read 2026.
`https://api.afl.com.au/`

Beaudouin, F., aus der Fünten, K., Tröß, T., Reinsberger, C. & Meyer, T. (2019), 'Head injuries in professional male football (soccer) over 13 years: 29% lower incidence rates after a rule change (red card)', *British Journal of Sports Medicine* **53**(15), 948–952.

Black, G. M. & Gabbett, T. J. (2014), 'Match intensity and pacing strategies in rugby league: an examination of whole-game and interchanged players, and winning and losing teams', *Journal of Strength and Conditioning Research* **28**(6), 1507–1516.

Cariboo, D. (2026), 'transfermarkt-datasets: football data from Transfermarkt', Data set, CC0; snapshot retrieved 3 August 2026, upstream commit af8c44a0.
`https://github.com/dcaribou/transfermarkt-datasets`

Cunnane, C. (1978), 'Unbiased plotting positions — a review', *Journal of Hydrology* **37**(3–4), 205–222.

David, H. A. & Nagaraja, H. N. (2003), *Order Statistics*, 3rd edn, Wiley, Hoboken, NJ.

Del Corral, J., Barros, C. P. & Prieto-Rodriguez, J. (2008), 'The determinants of soccer player substitutions: a survival analysis of the Spanish soccer league', *Journal of Sports Economics* **9**(2), 160–172.

Dilger, A. & Vischer, L. (2024), 'Effects of the rule change from three to five substitutions in the Bundesliga', *International Journal of Sport Finance* **19**(4), 198–207.

Dixon, M. J. & Coles, S. G. (1997), 'Modelling association football scores and inefficiencies in the football betting market', *Journal of the Royal Statistical Society: Series C (Applied Statistics)* **46**(2), 265–280.

Euroleague Basketball (2026), 'EuroLeague and EuroCup live data API', Public box-score interface, read 2026.
https://api-live.euroleague.net/

Fuller, C. W., Ekstrand, J., Junge, A., Andersen, T. E., Bahr, R., Dvorak, J., Hägglund, M., McCrory, P. & Meeuwisse, W. H. (2006), 'Consensus statement on injury definitions and data collection procedures in studies of football (soccer) injuries', *British Journal of Sports Medicine* **40**(3), 193–201.

Gaelic Athletic Association (2026), 'All-Ireland senior final match reports', Match reports and official team sheets, read 2026.
https://www.gaa.ie/

Gómez, M.-Á., Lago-Peñas, C. & Owen, A. L. (2016), 'The influence of substitutions on elite soccer teams' performance', *International Journal of Performance Analysis in Sport* **16**(2), 553–568.

Harrington, J. (2026), 'Football Expert Advisory Group proposals will require very high threshold for change', gaa.ie, 30 September, read 16 October 2026.
https://www.gaa.ie/article/football-expert-advisory-group-proposals-will-require-very-high-threshold-for-change

Hills, S. P., Barrett, S., Feltbower, R. G., Barwood, M. J., Radcliffe, J. N., Cooke, C. B., Kilduff, L. P., Cook, C. J. & Russell, M. (2019), 'A match-day analysis of the movement profiles of substitutes from a professional soccer club before and after pitch-entry', *PLOS ONE* **14**(1), e0211563.

Hirotsu, N., Masui, Y., Shimasaki, Y. & Yoshimura, M. (2024), 'Modelling tactical changes in association football using a Markov game',

IMA Journal of Management Mathematics **35**(3), 503–535. Correction: IMA Journal of Management Mathematics, 36(3), 615, 2025, doi 10.1093/imaman/dpaf014.

Hirotsu, N. & Wright, M. (2002), 'Using a Markov process model of an association football match to determine the optimal timing of substitution and tactical decisions', *Journal of the Operational Research Society* **53**(1), 88–96. Erratum: Journal of the Operational Research Society, 53(10), 1174, doi 10.1057/palgrave.jors.2601414.

Hirotsu, N. & Wright, M. (2005), 'Modelling a baseball game to optimise pitcher substitution strategies incorporating handedness of players', *IMA Journal of Management Mathematics* **16**(2), 179–194.

Hunter, J. D. (2007), 'Matplotlib: a 2D graphics environment', *Computing in Science & Engineering* **9**(3), 90–95.

It's Rugby (2026), 'itsrugby.fr', Match sheets of the Top 14, Pro D2, Premiership, United Rugby Championship and Super Rugby, read 2026.
`https://www.itsrugby.fr/`

Kadane, J. B. (2016), 'Sums of possibly associated Bernoulli variables: the Conway–Maxwell-binomial distribution', *Bayesian Analysis* **11**(2), 403–420.

LeagueEyeTest (2021), 'The decline of 80 minute players – NRL round 12 2021 stats and trends', Rugby League Eye Test, 1 June.
`https://www.rugbyleagueeyetest.com/2021/06/01/the-decline-of-80-minute-players-nrl-round-12-2021-stats-and-trends/`

Maher, M. J. (1982), 'Modelling association football scores', *Statistica Neerlandica* **36**(3), 109–118.

Maia, L. F. G. N., Pennanen, T., da Silva, M. A. H. B. & Targino, R. S. (2026), 'Stochastic modelling of football matches using dynamic regressors', *International Journal of Forecasting* **42**(2), 691–707.

Meyer, J. & Klatt, S. (2021), 'Impact of one additional substitution on player load and coaching tactics in elite football', *Applied Sciences* **11**(16), 7676.

MLB Advanced Media (2026), 'MLB stats API', Public statistics interface, game logs 2008–2025, read 2026.
`https://statsapi.mlb.com/api/v1/`

Mota, G. R., dos Santos, I. A., Arriel, R. A. & Marocolo, M. (2020), 'Is it high time to increase elite soccer substitutions permanently?', *International Journal of Environmental Research and Public Health* **17**(19), 7008.

Myers, B. R. (2012), 'A proposed decision rule for the timing of soccer substitutions', *Journal of Quantitative Analysis in Sports* **8**(1).

National Hockey League (2026), 'NHL web and statistics APIs', Public statistics interfaces, shift charts, read 2026.
`https://api-web.nhle.com/v1/`

National Rugby League (2026), 'NRL.com draw and match centre', Match-centre feed, seasons 2013–2025, read 2026.
`https://www.nrl.com/draw/`

Nevill, A. M., Smith, J. & Cloak, R. (2026), 'How have body size, age, and the use of substitutes evolved in English professional football between the 6 seasons 1973–4, 1983–4, 1993–4 to 2023–4?', *International Journal of Sports Science & Coaching* (online first, 3 June).

Pappalardo, L., Cintia, P., Rossi, A., Massucco, E., Ferragina, P., Pedreschi, D. & Giannotti, F. (2019), 'A public data set of spatio-

temporal match events in soccer competitions', *Scientific Data* **6**, 236.

Ribeiro, C. F. B., Siqueira, L. S., Pinto, D. P. & Silva, C. D. (2020), 'The three and six-substitution rules in football: a preliminary comparative analysis in quantitative replacing, game statistics, win rate and winning probability', *Motriz: Revista de Educação Física* **26**, e10200188.

Ricou, G. P. (2026), 'One equation allocates playing time in competitive team sports: the rulebook sets the arithmetic and the league sets the habit', arXiv:2609.05584v1 [physics.soc-ph], version 1, 4 September 2026.
`https://arxiv.org/abs/2609.05584v1`

Roberts, S. P., Stokes, K. A., Williams, S., West, S. W., Kemp, S., Cross, M., Moore, I. S., Bitchell, C. L., Mathema, P., Quarrie, K., McDonald, W., Fortington, L., Rubio del Castillo, E., Readhead, C., Sewry, N., Falvey, É. & Tucker, R. (2024), 'Injury in starting and replacement players from five professional men's rugby unions', *Sports Medicine* **54**(8), 2157–2167.

Shmueli, G., Minka, T. P., Kadane, J. B., Borle, S. & Boatwright, P. (2005), 'A useful distribution for fitting discrete data: revival of the Conway–Maxwell–Poisson distribution', *Journal of the Royal Statistical Society: Series C (Applied Statistics)* **54**(1), 127–142.

Silva, R. M. & Swartz, T. B. (2016), 'Analysis of substitution times in soccer', *Journal of Quantitative Analysis in Sports* **12**(3), 113–122.

Sun, R., Wang, C., Qin, Z. & Han, C. (2024), 'Temporal features of goals, substitutions, and fouls in football games in the five major European league from 2018 to 2021', *Heliyon* **10**(5), e27014.

The International Football Association Board (2020), 'Five substitutes option temporarily allowed for competition organisers', News release, 8 May.
https://www.theifab.com/news/five-substitutes-option-temporarily-allowed-for-competition-organisers/

The International Football Association Board (2022), 'The IFAB permanently approves five-substitute option in top-level competitions', News release, 13 June; Laws of the Game 2022/23, Law 3.2.
https://www.theifab.com/news/the-ifab-permanently-approves-five-substitute-option-in-top-level-competitions/

Titman, A. C., Costain, D. A., Ridall, P. G. & Gregory, K. (2015), 'Joint modelling of goals and bookings in association football', *Journal of the Royal Statistical Society Series A: Statistics in Society* **178**(3), 659–683.

Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D., Burovski, E., Peterson, P., Weckesser, W., Bright, J., van der Walt, S. J., Brett, M., Wilson, J., Millman, K. J., Mayorov, N., Nelson, A. R. J., Jones, E., Kern, R., Larson, E., Carey, C. J., Polat, İ., Feng, Y., Moore, E. W., VanderPlas, J., Laxalde, D., Perktold, J., Cimrman, R., Henriksen, I., Quintero, E. A., Harris, C. R., Archibald, A. M., Ribeiro, A. H., Pedregosa, F., van Mulbregt, P. & SciPy 1.0 Contributors (2020), 'SciPy 1.0: fundamental algorithms for scientific computing in Python', *Nature Methods* **17**(3), 261–272.

Wang, Z., Veličković, P., Hennes, D., Tomašev, N., Prince, L., Kaisers, M., Bachrach, Y., Elie, R., Wenliang, L. K., Piccinini, F., Spearman, W., Graham, I., Connor, J., Yang, Y., Recasens, A., Khan, M., Beauguerlange, N., Sprechmann, P., Moreno, P., Heess, N., Bowling,

M., Hassabis, D. & Tuyls, K. (2024), 'TacticAI: an AI assistant for football tactics', *Nature Communications* **15**, 1906.

Wittkugel, J., Memmert, D. & Wunderlich, F. (2022), 'Substitutions in football – what coaches think and what coaches do', *Journal of Sports Sciences* **40**(15), 1668–1677.

World Rugby (2020), 'Rugby world cup 2019: statistical analysis', World Rugby Game Analysis.
`https://www.world.rugby/news/691225/rugby-world-cup-2019-statistical-analysis`

Wright, M. (2014), 'OR analysis of sporting rules – a survey', *European Journal of Operational Research* **232**(1), 1–8.

Wyscout (2026), 'Minutes played', Wyscout Glossary, read 15 October 2026.
`https://dataglossary.wyscout.com/minutes/`

Xiao, Z. & Zhang, H. (2024), 'More substitutions changed team substitution strategy? An analysis of the FIFA World Cup 2002–2022', *BMC Sports Science, Medicine and Rehabilitation* **16**, 165.